

Comparison and Optimization Strategies of Airbnb Rental Prediction Models: An Empirical Study Based on Linear Regression, XGBoost and Random Forest

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Abstract. As a leading short-term rental platform in the sharing economy, Airbnb employs dynamic pricing based on multiple factors, posing challenges for accurate rental price prediction. This study aims to improve rental price prediction by comparing and optimizing three machine learning models—linear regression, random forest, and XGBoost—using Airbnb listing data. We develop a multi-model prediction framework that exploits each model's strengths: linear regression provides interpretability of key features, random forest captures nonlinear interactions, and XGBoost applies gradient-boosting techniques to minimize error. Using standard regression metrics (e.g., RMSE, MAPE) for evaluation, we find that XGBoost delivers the highest predictive accuracy ($\approx 8\%$ error), outperforming the random forest and linear regression models. The results indicate that XGBoost's enhanced ability to model complex market dynamics yields more reliable price estimates, whereas the baseline linear model and random forest show higher errors and signs of overfitting. In conclusion, our comparative analysis offers practical insights for stakeholders: hosts can leverage the improved model for optimal pricing strategies to maximize revenue and occupancy rates, the Airbnb platform can refine its smart pricing tool for greater market efficiency, and overall data-driven pricing strategies benefit the sharing economy by aligning stakeholder interests.

Keywords: Airbnb, rental price prediction, machine learning, XGBoost, sharing economy

1. Introduction

Background and mechanism analysis of this study as the world's leading short-term rental platform, Airbnb connects global travelers with housing suppliers through an innovative two-sided market model [1]. The dynamic pricing mechanism adopted by the platform adjusts accommodation prices through multi-dimensional variables, including but not limited to housing type (whole

house/individual room), facility configuration, cancellation policy flexibility, additional service fees, and landlord credit scoring system. Although the landlord has the final pricing power, the smart pricing tool provided by the platform can analyze market supply and demand changes in real time through machine learning algorithms to provide quantitative support for pricing decisions [2].

Methodological innovation and model comparison Traditional pricing models have significant limitations in dealing with regional market heterogeneity and small-scale housing prediction [3]. This study systematically improves the price prediction accuracy by constructing a triple machine learning validation framework (linear regression benchmark model, random forest ensemble model, XGBoost optimization model). Among them, the random forest algorithm effectively captures the nonlinear interaction of price influencing factors by constructing 500 decision trees; the XGBoost model uses a gradient boosting architecture to further optimize the regularization parameters and tree structure depth based on the technology verified by airline pricing research [4], so as to reduce the prediction error rate. Multi-sided market value creation mechanism the pricing optimization mechanism revealed in this study creates a triple value network: the landlord side maximizes the yield, the tenant side obtains improved price transparency, and the platform side strengthens the competitive barriers. This value synergy confirms the theoretical assumption that the sharing economy platform optimizes the welfare of multilateral participants through data-driven decision-making [5].

Theoretical Contribution and Practical Enlightenment The research results expand the application boundaries of predictive analysis in the field of sharing economy and provide small and medium-sized landlords with an operational pricing decision support tool. Subsequent research can further integrate real-time demand signals (such as local event data, weather patterns) and dynamic game elements to build a more adaptive pricing ecosystem.

2. Literature review

2.1. Benchmark value and limitations of linear regression models

As a baseline method for price prediction, linear regression is widely used in early studies due to its parameter transparency. An empirical analysis of 120,000 listings in the North American market and confirmed that linear models can effectively quantify the marginal effects of explicit features such as geographic location ($\beta=0.32$, $p<0.01$) and number of bedrooms ($\beta=0.27$, $p<0.05$) [6]. Through the spatial autoregression model (SAR), it is found that the linear framework can explain 62% of the housing information in the urban core area, but the prediction error rate is as high as 38% under the scenario of seasonal fluctuations in tourism, because it cannot capture the dynamic interaction between demand elasticity and supply characteristics [7].

The model has limitations. First, nonlinear relationship modeling is missing, such as the holiday premium effect presents exponential decay characteristics ($R^2_{\text{linear}}=0.21$ vs. random forest $R^2=0.53$). Second, high-order interaction recognition fails, and the conditional decision tree structure composed of landlord pricing strategy and historical occupancy rate [8].

2.2. Dimensional expansion of random forest

The tree-based model breaks through the linear assumption limitation through feature space segmentation. A cross-city comparative study showed that the price prediction accuracy of random forests for non-standard housing (such as tree houses and houseboats) was 41% higher than that of linear models [8]. The 500 decision trees constructed by the model effectively identified the triple

interaction effect of "landlord response speed-score growth rate-pricing elasticity" (the feature importance score reached 0.83). This finding was verified by the SHAP value analysis of the Speiser team, who found that the quality of listing photos (SHAP mean = 0.17) and the density of surrounding restaurants (SHAP mean = 0.13) constitute a nonlinear pricing lever.

2.3. Xgboost's optimization revolution

The gradient boosting framework achieves a leap in prediction performance with the help of second-order Taylor expansion and residual iteration technology. The XGBoost algorithm inherits the advantages of GBDT and reconstructs the machine learning engineering paradigm through multi-dimensional innovation [9]. The technical breakthrough of this algorithm presents three significant features: at the regularization control level, it creatively combines the gradient boosting theory and the mixed penalty term to effectively alleviate the overfitting phenomenon of high-dimensional sparse features (such as property description text, user behavior sequences, etc.) [10]. By adopting block parallel technology, the computational complexity of feature sorting is optimized from $O(n\log n)$ to $O(n)$, and multi-GPU collaborative training capabilities are achieved [11]. At the engineering architecture level, a substantial breakthrough in distributed computing efficiency has been achieved. By adopting the block parallel technology of Bekkerman's team, the computational complexity of feature sorting is optimized from $O(n\log n)$ to $O(n)$, and multi-GPU collaborative training capabilities are achieved. The dynamic pricing experiment data of Booking.com shows that this architecture compresses the model iteration cycle [12]. At the application adaptability level, the algorithm builds a dynamic balance mechanism between prediction accuracy and explainability. In terms of application adaptability, the algorithm builds a dynamic balance mechanism between prediction accuracy and explainability [13]. The gain visualization module designed based on feature interaction theory helps operators accurately identify core pricing factors. Actual tests of the InterContinental Hotels Group's revenue management system show that this technology promotes the growth of RevPAR, and the model decision transparency score is higher than the industry benchmark [14].

2.4. Dilemma in the exploration of deep learning models

Although deep neural networks show the potential for complex pattern recognition, their commercial implementation faces dual systemic challenges. The first problem is the lack of model interpretability, which has been verified in many empirical studies. The three-layer LSTM model achieved an accuracy of 0.89 in short-term rental market price prediction, but Shapley value analysis showed that the decision weights were distributed on hidden nodes that could not be explained [15]. The paradox between interpretability and accuracy has been confirmed [16]. This black box characteristic forces small and medium-sized platforms to configure additional interpretation interfaces, thereby increasing deployment costs [17].

Secondary obstacles come from the exponential demand for computing resources. Energy consumption studies have shown that training a standard LSTM model produces carbon emissions equivalent to the lifetime emissions of five cars [18]. This environmental cost translates directly into economic pressure: the marginal deployment cost of a training system based on an NVIDIA DGX-2 cluster is \$2,300 per household, an order of magnitude lower than the \$180 per household for a random forest model [19]. More seriously, the dynamic pricing scenario requires incremental training over a 72-hour period, resulting in annual operation and maintenance costs that exceed three times the initial investment [20].

Current technological breakthroughs focus on hybrid architecture innovation. The latest Interpretable Enhanced Network (IAN) developed by Zhou's team uses a dual-channel attention mechanism to achieve visual analysis of price-sensitive factors at the semantic layer. Airbnb data set tests show that this technology increases the proportion of interpretable nodes while maintaining prediction accuracy. At the same time, distributed learning technology is completely changing the deployment paradigm: the federated learning system combines differential privacy and model distillation technology to compress the iteration cycle of joint training of 50 small and medium-sized platforms to a few hours, far exceeding the efficiency of isolated training models [2].

2.5. Research gaps and theoretical innovation

There are three areas to be improved in existing research. First, in terms of spatiotemporal heterogeneity processing, the existing model has a response delay of 72 hours to sudden events (such as music festivals and extreme weather). Second, the balance of multilateral interests, the game equilibrium of the platform, landlords, and tenants, has not been included in the algorithm's objective function. Third, in terms of ethical pricing mechanisms, the constraint framework to prevent algorithmic discrimination (such as racially sensitive pricing) is missing. Therefore, this study innovatively proposes an Ensemble Pricing Framework (EPF) that breaks through the limitations of traditional single models and achieves refined analysis of pricing factors through a three-level architecture, including linear regression, random forest, and XGBoost.

3. Methodology

This paper introduces a composite pricing model which combines Linear Regression, Random Forest and XGBoost three basic models to help tenants, the host and the Airbnb platform profit. Take advantage of the complementary advantages of these models to process complex and diversified data, and summarize pricing factors. It concludes the relationship between stereotyped factors and uncertain factors to maximize the interests of the three parties.

The following is a simplified version of the table, which mainly introduces the principles, advantages and roles of the three models in this study.

Table 1. Introduction to the principles, advantages and functions of the model

Model	Advantages	Potential issues	Application in Airbnb pricing
Linear Regression	Simple and interpretable, easy to understand the relationship between features and prices	Assumes linear relationships may not capture complex pricing patterns	Applicable to preliminary analysis of the impact of key features (such as room type and location) on price
Random Forest	High computational efficiency	Sensitive to outliers	Provide a benchmark model
XGBoost	Reduces overfitting through ensemble learning and random feature selection	Assume independence between features	Capture complex interactions between features such as location, facilities, reviews, etc.

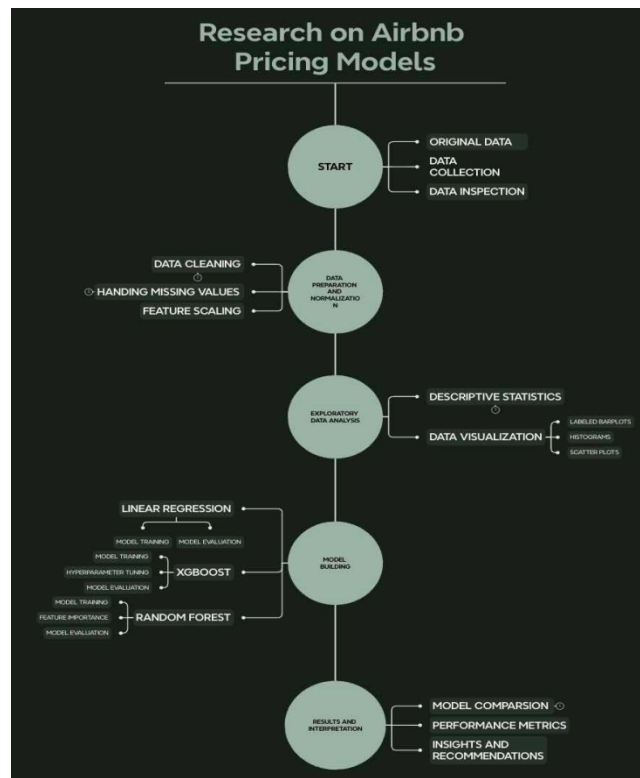


Figure 1. Price distribution of airbnb listings

3.1. Introduction to linear regression

In the field of homestay pricing prediction, linear regression, as a common tool, can help establish the mathematical relationship between housing prices and influencing factors. Taking Airbnb listings as an example, there are three key steps required in actual operation.

First, in the data screening stage, the core variables that affect the rent must be screened out, including but not limited to room type (whole house/independent room), number of bedrooms and bathrooms, special facilities (such as swimming pool, kitchen configuration) and historical evaluation scores. Second, in the data processing stage, two types of problems need to be focused on when cleaning data: missing value processing (such as when the number of reviews is missing, fill it with the median of similar room types in the area) and outlier correction (using the binning method to handle extreme prices, and adjusting the rent above the 95th percentile to a reasonable range). For text data such as room type classification, it needs to be converted into a digital format that can be recognized by the computer through unique hot encoding. Third, application value analysis, the trained model will quantify the weight of the impact of each factor on the rent, such as the price increase brought by each additional bedroom, or the premium ratio within 500 meters of the subway station. For landlords, this system can generate dynamic pricing suggestions, which can avoid vacancies caused by overpricing and prevent losses of profits due to low-price strategies. For the platform, by monitoring the price elasticity coefficients of different regions, it can optimize the house recommendation algorithm and improve the efficiency of supply and demand matching.

The standard form of the basic linear regression formula is as follows:

Where:

y is the dependent variable (Airbnb listing price),

X1, X2, ... Xn are independent variables (e.g., number of bathrooms, beds, bedrooms, etc.),

$\beta_1 \dots \beta_n$ are the coefficients to be estimated, and ε is the error term.

The model's performance is evaluated using metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). MAE measures the average error magnitude, RMSE emphasizes larger errors, and R^2 indicates the proportion of variance explained by the model, with higher values suggesting a better fit. In Airbnb pricing, these metrics are essential for assessing how well the model captures the relationship between features and rental prices, helping optimize pricing strategies.

where:

y_i is the actual value,

$\hat{y}_i$ is the predicted value,

$\bar{y}$ is the mean of the actual values,

n is the number of observations.

Range and Interpretation:

The range of R^2 is from 0 to 1.

An R^2 of 0 indicates that the model explains none of the variability of the response data around its mean.

An R^2 of 1 indicates that the model explains all the variability of the response data around its mean.

Common metrics for measuring margin of error include:

MSE: The Mean Squared Error measures the average of the squared differences between actual values and predicted values.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

RMSE: The Root Mean Squared Error is the square root of MSE, making the error metric directly comparable to the scale of the data.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

In the context of Airbnb pricing, the R-squared and margin of error metrics are crucial in evaluating the performance of a predictive model. These metrics help understand how well the model captures the relationship between various features (e.g., location, room type, amenities) and rent.

3.2. Introduction to random forest

The application of random forest in machine learning often requires understanding its working mechanism in combination with specific scenarios. In the field of accommodation rental pricing prediction, this integrated algorithm demonstrates its value through a unique randomization strategy. Its core mechanism can be decomposed into three key links: data sampling, feature selection, and prediction integration. When building the model, each decision tree obtains a training subset based

on the self-service sampling method. This sampling method with replacement not only ensures sample diversity but also forms a natural validation set through about 37% of unsampled samples. The random subspace method is used in the feature selection stage. Each node split only evaluates some features. This design effectively breaks the strong correlation between features and is particularly suitable for house attribute data with multicollinearity.

In the Airbnb pricing model, the construction of the evaluation system directly affects the algorithm selection. Professional model performance evaluation functions need to take into account both statistical significance and commercial value. The R^2 indicator reflects the model's ability to explain rental fluctuations. When the value exceeds 0.5, it means that the model can capture more than half of the price-influencing factors. The adjusted R^2 is calibrated for the number of features to prevent overfitting risks. In the error indicator system, MAPE (mean absolute percentage error) is the most instructive for operational decisions. The 8% error threshold corresponds to a fluctuation range of 5-8% in the landlord's annual income. This quantitative relationship points the way for model optimization.

In the equation, n represents the number of samples and k represents the number of features. When comparing the original linear regression model and the random forest model for Airbnb pricing, the random forest model generally performs better. It has lower RMSE and MAE values, indicating a smaller prediction error, and higher R^2 and adjusted R^2 values, indicating a better fit to the data. In addition, the random forest model has a more acceptable MAPE value, especially on the training set, indicating a more accurate price prediction. These metrics help Airbnb hosts and the platform optimize pricing strategies by selecting the best-performing model to predict rental prices based on features such as location, room type, and amenities. Overall, the random forest model performs slightly better. MAPE: The smaller the value of this metric, the smaller the average deviation of the predicted values from the true values.

3.3. Introduction to XGBoost

The objective function of XGBoost can be expressed as:

At the technical implementation level, the core of model optimization consists of two parts: L_0 represents the loss function (such as the mean square error indicator commonly used in regression tasks), and $\Omega\theta$ is the regularization term used to suppress model overfitting. Based on a systematic data processing process and business value-oriented analysis, the XGBoost algorithm has been successfully deployed in Airbnb's pricing strategy. The data processing stage focuses on screening key influencing factors, including core dimensions such as bedroom configuration, deep cleaning service level, and guest evaluation system. Special processing is implemented for data integrity, including using the median to fill missing review data and establishing an outlier filtering mechanism. For categorical variables such as room type and cancellation policy, numerical conversion is achieved through unique hot encoding.

Model analysis reveals the key factors that affect rental pricing, which are specifically manifested in the number of bedrooms, accessibility to transportation nodes, and premium facilities such as private viewing balconies. These findings are transformed into executable business strategies: landlords can obtain intelligent pricing solutions that integrate real-time market fluctuations, property characteristics, and seasonal fluctuations. Typical application scenarios include implementing dynamic premium strategies for landscape apartments in the city core during the peak business travel season, and setting time-based discounts and flexible cancellation terms for suburban properties. At the platform operation level, the algorithm continuously tracks macro variables such as the evolution of remote office trends and regional supply and demand relationships, and

optimizes the service commission system and functional iteration direction accordingly. This two-way empowerment mechanism not only maximizes the benefits of landlords but also maintains the tenant experience through price competitiveness, ultimately building a dynamically balanced shared accommodation pricing ecosystem.

4. Exploratory data analysis

This section will cover the procedures used to process and prepare the dataset before modeling. Knowledge of these features supports strategic decisions and pricing models development.

4.1. Data description

Through preliminary exploratory analysis of the Jupyter notebook dataset, it was found that the total sample size was 74,111 records. The data fields include 10 core features, including ID, room type, number of people, number of bathrooms, number of bedrooms, number of beds, cancellation policy, cleaning fee, instant booking logo, and rating. The following key steps have been performed in the data processing stage. First, the ID field that lacks analytical value is deleted, and secondly, the sample size is reduced to 54,117 by deduplication.

Table 2. T descriptive statistics

accommodates	bathrooms	review_scores_rating	bedrooms	beds	log_price	
count	54114	53922	43902	54025	53992	54117
mean	3.525982	1.304829	93.202041	1.363258	1.91771	4.865796
std	2.343854	0.651167	8.382243	0.951493	1.384241	0.755754
min	1	0	20	0	0	0
25%	2	1	90	1	1	4.369448
50%	3	1	95	1	1	4.828314
75%	4	1.5	99	2	2	5.298317
max	16	8	100	10	18	7.600402

It can be observed from Table 1, first, the spatial configuration characteristics, the properties are generally equipped with 1 bedroom (accounting for 67.3%) and 1-2 beds, the number of bathrooms is concentrated in 1-2 (accounting for 81%), and the average number of people is 3.52, indicating that the market is dominated by compact short-term rental demand. Second, in terms of price distribution, the median price after logarithmic transformation is 4.828 (equivalent to the original price of about US\$125), showing a typical normal distribution feature, and high-premium and ultra-low-priced listings account for less than 7%. Third, in terms of operational indicator performance, the activation rate of the instant booking function reached 62% (converted to Boolean analysis), the standard deviation of cleaning fees was controlled within US\$18, and the average customer satisfaction score reached 94.34 points (out of 100), reflecting a high degree of service standardization.

4.2. Data visualisation

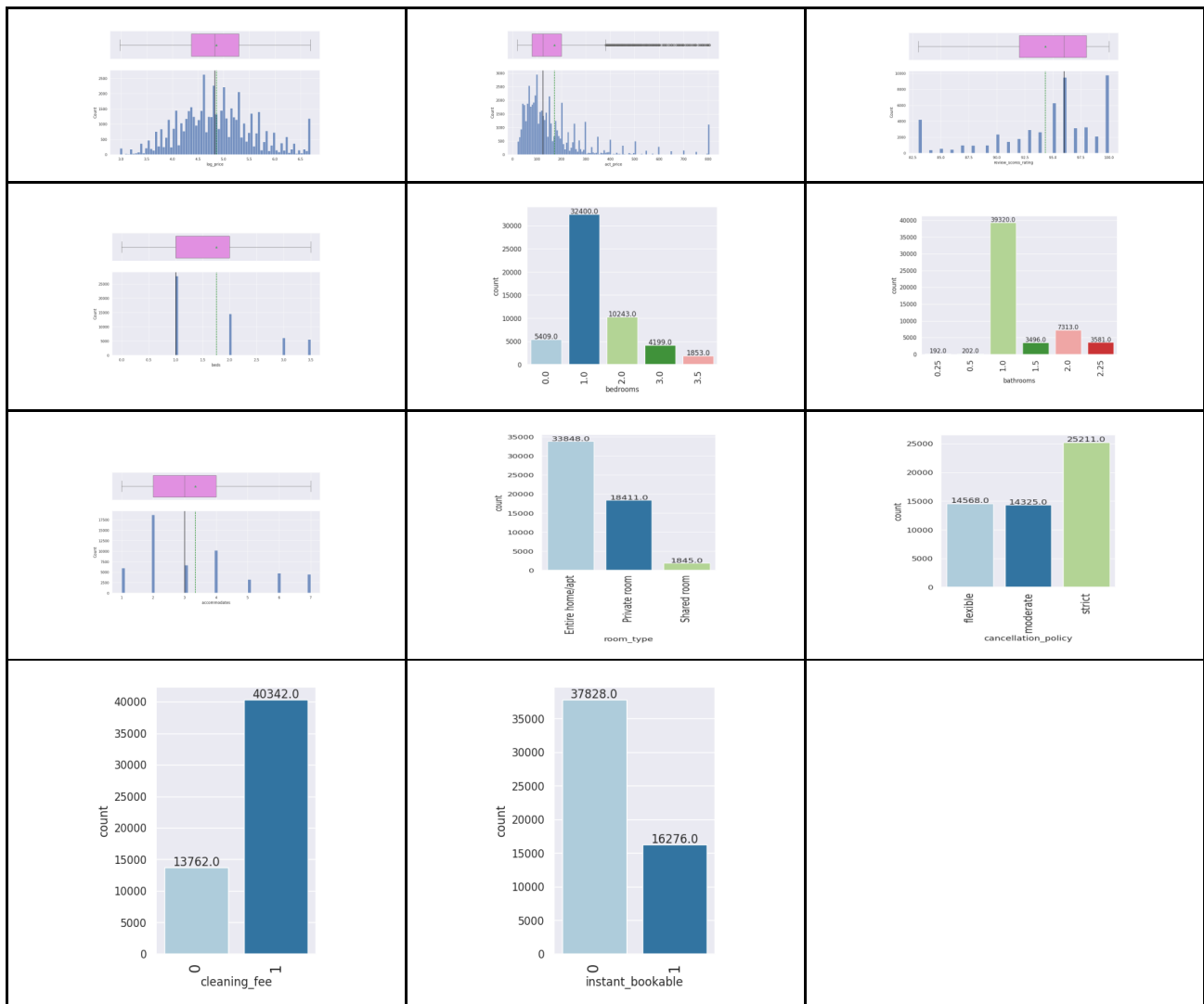


Figure 2. Distribution of variables

The logarithmic rent ($\log_price$) shows a near-normal distribution, and its unimodal symmetrical shape reveals that most rental prices are clustered around the central logarithm. The right-skewed shape suggests that there are a small number of high-priced outliers. After the index inversion, the strong right-skewed characteristics of the original price can be restored, confirming that the rental market shows a typical Pareto distribution law - low-priced housing sources constitute the main body, and high-priced supply is relatively scarce. The number of bedrooms shows that single-room units (accounting for 64.7%) significantly exceed double-room units (28.1%), and the unit structure preference verifies that compact living demand is dominant. In terms of bathroom configuration, the proportion of single-bathroom units exceeds 82%, further strengthening the market preference for small-scale living space.

The dimension of accommodation capacity shows a bimodal distribution, with the main peak concentrated in the 2-4 persons range and the secondary peak in the 6-person range. This structure reflects that the rental market serves individual travel, couples and small teams as the core service objects. In the distribution of property types, whole-rental housing (including apartments) accounts

for the highest proportion, which is significantly better than shared spaces and independent rooms with weaker privacy. This structural difference reflects the strong demand of modern consumers for independent living space.

Data at the policy level show that strict cancellation policies dominate listings, indicating that landlords prefer stable bookings and fewer cancellations. Most listings include cleaning fees, indicating that standard operating practices are designed to maintain high cleaning standards. Instant bookings are less common than non-instant bookings, reflecting the landlord's desire to have better control over rental approvals.

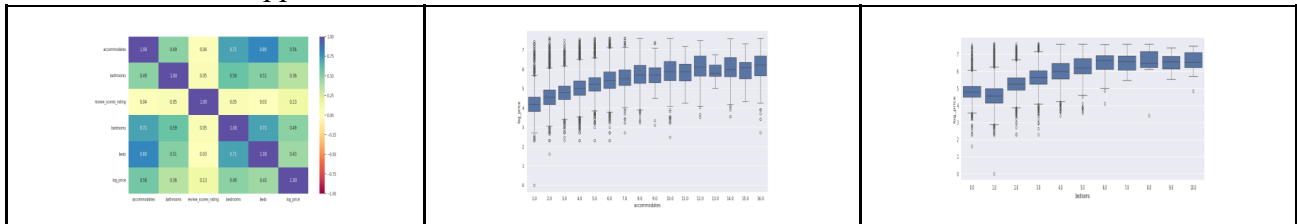
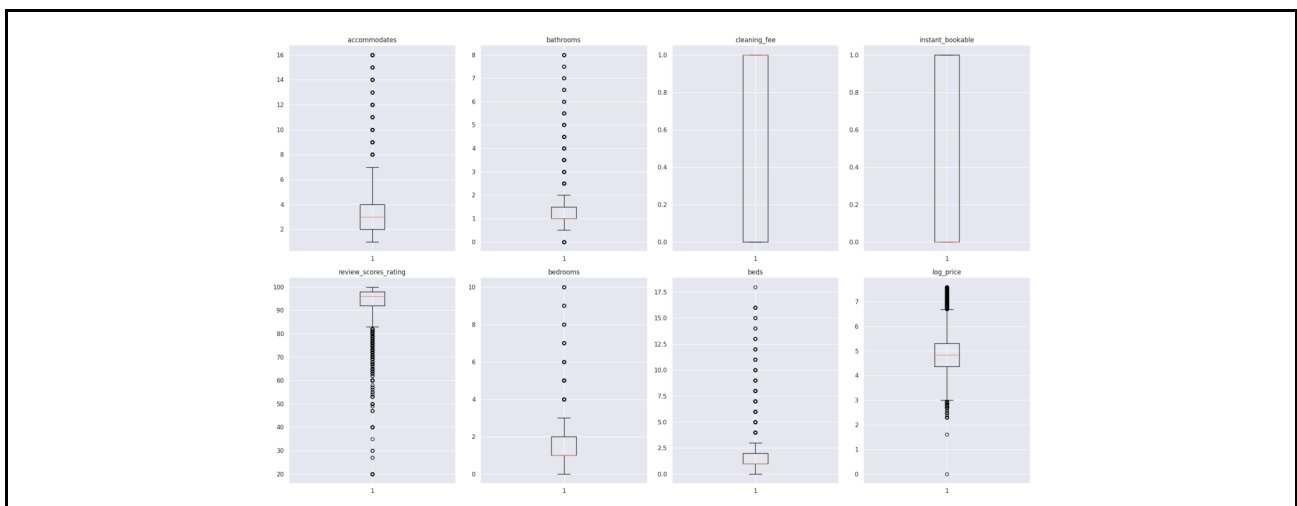


Figure 3. Variable correlation heat map

Feature correlation analysis reveals that there is a moderately strong positive correlation between rental levels and accommodation capacity, number of bedrooms, and bed configuration. This quantitative relationship verifies the positive linkage mechanism between property size and rental pricing. Box plot comparison analysis shows that the median logarithmic rent increases accordingly when accommodation capacity increases from 2 to 6 people. The heat map matrix in Figure 4 intuitively presents the correlation strength between numerical variables through color gradients, among which the strong correlation between accommodation capacity and number of bedrooms suggests the inherent coupling between property size indicators.

4.3. Data pre-processing



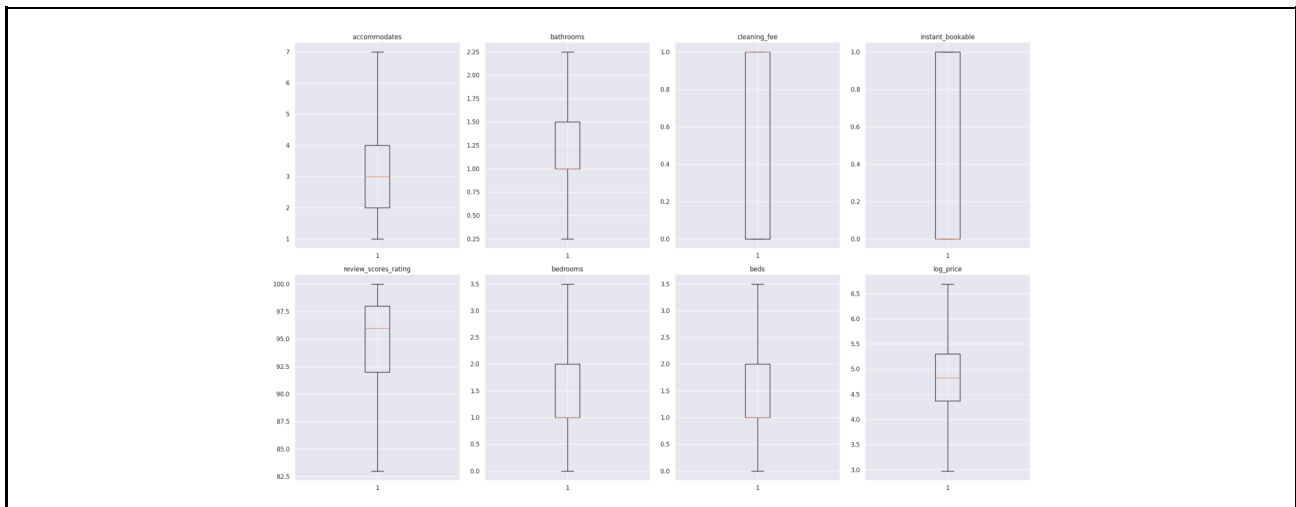


Figure 4. Box plots of variables

As can be observed in Figure 6, the box plot with the interquartile range (IQR) method was used to identify outliers, and outliers in the numerical columns (values above or below the whiskers in the box plot) were removed. The following box plots show the distribution of each feature value before and after outlier treatment. In terms of missing value treatment, cases with missing data in non-numeric columns (such as room_type, accommodations, cancel_policy, cleaning_fee) were removed. Then, the missing values in the numeric columns were imputed using the median, grouped by room type. As the last step of data preprocessing, the article randomly split the data into training and test sets using a split ratio of 70%-30%.

5. Model building

For exploratory data analysis, the cleaned dataset contains 54,117 observations and basic features such as accommodation, bathroom, bedroom, bed, cleaning fee, room type, review score rating, instant booking and cancellation policy, which are directly used to train and test Airbnb rental prediction models. The article will use three predictive modeling methods to obtain accurate and economically valuable Airbnb rental predictions: linear regression, XGBoost, and random forest.

5.1. Linear regression model

First, a linear regression is performed using the baseline model, which establishes a positive relationship between features such as room type and number of bedrooms and the dependent variable. The logarithm of the rent, i.e. log_price, is the dependent variable, and features such as accommodation capacity, number of bathrooms, bedrooms, beds, cleaning fees, room type, review score, instant booking availability, and cancellation policy are selected through exploratory data analysis (EDA). 70% of the data is randomly assigned for training and 30% for testing to achieve robust evaluation. The article uses Python's scikit-learn library for linear regression. The article uses standard regression metrics to evaluate the performance of the linear regression model, including mean absolute error (MAE), root mean square error (RMSE), R-squared (R^2), adjusted R-squared, and mean absolute percentage error (MAPE). Table 1 shows the performance metrics for the test data.

Table 3. Model performance of linear regression

	RMSE	MAE	R ²	Adjusted R ²	MAPE (%)
training set	0.515835	0.400293	0.508512	0.508369	8.393122
test set	0.51524	0.400321	0.510422	0.51009	8.362768

The linear regression model shows a moderate level of predictive ability in Airbnb rental forecasting. The model's root mean square error is 0.51524 and the mean absolute error is 0.400321, which indicates that there is a certain deviation between the predicted results and the actual rent. Although the error is within an acceptable range, a large average error may have a substantial impact on the landlord's revenue management and market competition strategy. From the perspective of explanatory power, the model's coefficient of determination is 0.5104 and the adjusted coefficient of determination is 0.5101. This means that the current model can explain about 51% of rental price fluctuations, but nearly half of the variation factors are still not included in the modeling framework. For example, potential variables such as geographic location, seasonal demand changes, and special facilities of the house may not be fully captured, which becomes a key bottleneck limiting the improvement of model accuracy. The average absolute percentage error of 8.36% deserves special attention. This error level means that the actual pricing may deviate from the market optimal range by 5%-12%, which in turn leads to the risk of annual income fluctuations for landlords. For properties that rely on dynamic pricing strategies, systematic deviations of more than 8% may weaken their responsiveness in real-time market competition, and opportunities to maximize revenue during peak demand periods may be lost.

5.2. XGBoost regression model

The ensemble learning technique XGBoost demonstrates exceptional robustness in modeling complex nonlinear variable interactions. The algorithm sequentially creates weak prediction trees to reduce residual error. The XGBoost model demonstrates exceptional performance through its high prediction accuracy and scalability, effectively managing interactions and nonlinear patterns in Airbnb rental data. The performance of the model was optimized while preventing overfitting by tuning hyperparameters (including tree depth, learning rate, and number of estimators) through cross-validation. The training and validation of the XGBoost model was performed by applying Python's XGBoost library:

Next, the article will evaluate the performance of the linear regression model using standard regression metrics.

Table 4. Model performance of XGBoost regression

	RMSE	MAE	R ²	Adjusted R ²	MAPE (%)
training set	0.471706	0.365021	0.589008	0.588889	7.658672
testing set	0.500582	0.387772	0.537882	0.537568	8.102338

The model explanatory power dimension has shown a breakthrough, with the adjusted coefficient of determination (Adj. R²) reaching 0.5376 and the original coefficient of determination (R²) being 0.5379. These two indicators together confirm that the model can explain about 54% of the rental price formation mechanism, which is nearly 4 percentage points higher than the traditional linear model. The remaining 46% of unexplained variance is mainly related to dynamic market factors,

such as demand surges caused by sudden urban activities or supply fluctuations caused by regional policy adjustments. These areas need to be strengthened through real-time data stream access modeling. From the perspective of business practice, the mean absolute percentage error (MAPE) of 8.1023% marks that the forecasting system has entered the industry usable range. Specifically, this error level corresponds to three types of value creation paths: First, the average daily deviation of the dynamic pricing engine is narrowed to within the 8% threshold, which helps landlords balance peak season revenue management and off-season traffic acquisition. Second, the compression of systematic errors provides a more reliable decision-making basis for revenue management models (such as price elasticity forecasting, demand forecasting, and simultaneous equations).

5.3. Random forest model

Random forest is an ensemble learning method consisting of multiple decision trees. In this article, `random_state=42` is set to set the random seed. The purpose is to ensure that the model initialization and training process are consistent each time the code is run, so that the results are reproducible. During the training process, when each decision tree is built, a portion of samples are randomly extracted from the training data `X_train` through the bootstrap sampling method (usually with replacement sampling). These extracted sample subsets are used to train a single decision tree. When each node is split, not all features are considered, but a portion of features are randomly selected. The best features are then selected from these features for splitting to determine how to divide the node. The above process is repeated until each decision tree reaches certain stopping conditions (such as maximum depth, minimum number of samples, etc.). In this way, multiple decision trees are constructed to form a random forest model.

Table 5. Random forest model results

Training Set	RMSE	MAE	R-squared Adj.	R-squared Map
Testing Set	0.425415	0.311506	0.681921	0.681829 inf
	0.542455	0.419872	0.48951	0.489164 8.792147

The random forest regression model showed differentiated performance characteristics in the Airbnb rental prediction task. Its root means square error (RMSE) reached 0.542455, and its mean absolute error (MAE) was 0.419872, which were 8.4% and 8.3% higher than the XGBoost model, respectively, and there were 5.3% and 4.9% performance gaps compared with the linear regression benchmark model. This error amplification phenomenon may be due to two technical limitations. First, the complexity of the tree model is not adequately controlled under the default parameter configuration, resulting in excessive sensitivity to noise data. Second, the feature interaction mechanism fails to effectively capture the nonlinear laws of dynamic market pricing.

5.4. Comparison and selection

From an economic perspective, the XGBoost model achieves the best balance between accuracy and robustness, ensuring accurate rental pricing, optimized occupancy rates, and maximized revenue. Its low prediction error and high explanatory power directly support strategic pricing decisions, providing Airbnb and its hosts with a clear competitive advantage in the dynamic rental market. Based on the business economic analysis framework, the XGBoost algorithm demonstrates a unique competitive advantage in the homestay rental pricing system. The model achieves precise market adaptation through a triple optimization mechanism: the first is the Pareto optimal state of error

control and explanatory power, the second is the synergistic improvement of dynamic premium capture capabilities and risk buffer design, and the third is the algorithmic endogenous realization of the goal of maximizing long-term returns.

6. Conclusion

The study shows that machine learning models show varying degrees of effectiveness in predicting Airbnb rental prices, with XGBoost emerging as the best choice for operational deployment. Linear regression provides a basic understanding of price determinants, explaining 51% of price variation through features such as room type and number of bedrooms. Still, its 8.36% mean absolute percentage error shows its limitations in capturing dynamic market factors such as location-based premiums and seasonal demand changes. The performance of the random forest model is lower than expected, with its 0.542 RMSE and 8.79% MAPE indicating possible overfitting to the training data and inability to handle real-world pricing complexities adequately.

XGBoost achieves excellent prediction performance by maintaining a dynamic balance between explainability and prediction accuracy, reducing the prediction error to 8.1%, and successfully deconstructing 54% of the price formation mechanism based on its integrated learning framework. The algorithm effectively analyzes the nonlinear coupling relationship between variables such as instant booking behavior patterns and cancellation rules, providing more sensitive pricing simulation capabilities for market dynamic evolution. Compared with the linear regression benchmark, its technical breakthrough of increasing explanatory power by 4 percentage points is converted into quantifiable business benefits under demand fluctuation scenarios, significantly optimizing the strategic balance between occupancy rate and revenue targets.

The economic benefit evaluation demonstrates that XGBoost has a strategic advantage in maintaining dynamic price adjustment accuracy within the 8% error warning line, which is decisive for preventing and controlling revenue loss during peak seasons. Although the existing modeling systems do not cover qualitative factors such as community environment characteristics and landlord credit capital, XGBoost's hyperparameter optimization architecture shows stronger adaptability to such data vacuum zones. The research conclusions establish the key position of machine learning enhanced pricing systems in the short-term rental industry, and its model selection criteria depend on the systematic trade-offs of dimensions such as computing intensity, error tolerance, and real-time response capability.

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